SENSOR TECHNIQUES FOR DETECTING AND MONITORING MARINE LIFE

1. IDEA INTRODUCTION:

Tidal energy is a renewable energy that harnesses the power of ocean tides to generate electricity. It is an eco-friendly energy resource since it doesn't release any greenhouse gases. In fact, it has greater efficiency than solar or wind energy. There is scope to generate this energy on a large scale but there are many hindrances like frozen sea, low or weak tides, straight shore lines. Apart from this there is an ecological disturbance in marine life. In our project, we will develop a chemical discharger which will release 'DISTURBANCES CUES' a signal(chemicals) which marine animal will detect and will stay away from the tidal plant. This device can be put into the water body where the tidal plant is setup. This project is an attempt to understand the response mechanism of aquatic life and also solve one of the major issues hindering the setup of tidal plant. It also aims to conserve and sustainably use the oceans, seas and marine resources for sustainable development.

2. BLOCK DIAGRAM:

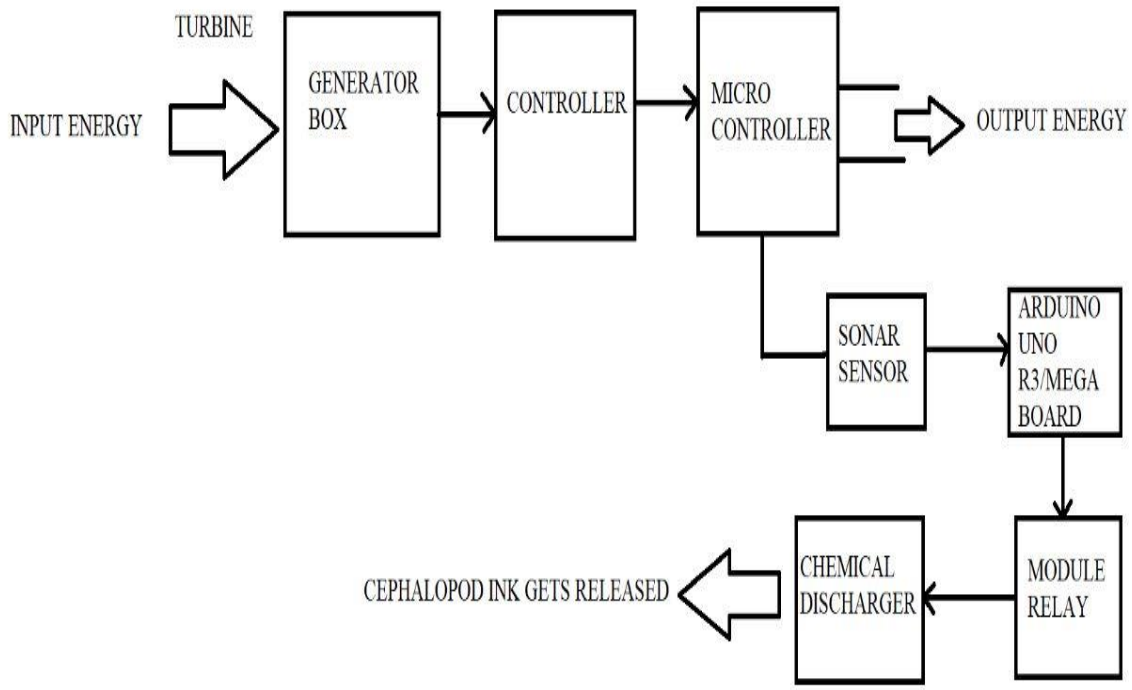

Fig.2.1 Block Diagram of Marine Life Protection System

3. HARDWARE DETAILS:

3.1 SONAR SENSOR:

The SONAR (Sound Navigation and Ranging) can detect and display the distribution, density, and movement of a school of fish at an angle of 360° or 180° in all directions. The figure below shows the SONAR Sensor:

Fig. 3.1 SONAR Sensor

3.2 ARDUINO UNO/ Mega Board:

ARM Cortex M33 with Trust zone / 160MHz / Ultra-low power MCU with FPU/ higher Security microcontroller or Arduino microcontroller board based on ATMega328P may be used for the application development. It has 14 input and output pins. It is used to establish a connection between SONAR Sensor and Module Relay. The figure below shows the Arduino Uno R3 microcontoller.

Fig.3.2 Arduino Microcontroller

3.3 12V MODULE RELAY:

A relay module is mainly used to switch electrical devices and systems ON or OFF by receiving electrical signals. The module relay receives the signal from the sensor and directs the Chemical Discharger to release the chemical. The following picture shows the 12V Module Relay:

Fig.3.3 12V Relay Module

3.4 CHEMICAL DISCHARGER:

The Chemical Discharger is used to release chemicals (in our project, Cephalopod Ink or Disturbance Cues). Cephalopod Ink is ejected out by the Cephalopod species when they encounter any danger. It also alerts other animals to move away from the danger zone. On the other hand, Disturbance Cues is a chemical released out by fishes when they sense danger. With respect to the availability, feasibility and eco-friendliness any one of the chemicals can be used. We will find out which one can be synthesised manually in order to protect marine life.

3.5 GENERATOR:

Tidal Generators are systems that generate electricity by capturing the energy of moving water with the tides. They use turbines that are fixed to the sea floor. The turbine rotates as the tides rise and fall. The following picture represents Tidal Turbine and Tidal Generator:

Fig.3.4 Tidal Turbine

Fig.3.5 Tidal Generator

3.6 SRF02 ULTRASONIC SENSOR:

An Ultrasonic Sensor is used to calculate the distance between two objects. The range of this sensor is 1.5cm to 6m. In our project, we will use this sensor to calculate the distance between the Tidal Plant and the nearby fish. It will help us to understand the change of concentration per cubic meter of the Cephalopod Ink and how it influences the response mechanism of fishes. The following picture represents the SRF02 Ultrasonic Sensor:

Fig.3.6 SRF02 Ultrasonic Sensor

3.7 WAVE-RIDING GENERATOR:

Wave riding generators are used to generate artificial waves. In our project, since we are demonstrating Tidal Energy Power Plant, we are using wave riding generators to show the same. The following picture represents Wave-Riding Generators:

Fig.3.7 Wave-Riding Generators

4 PROCESS FLOWCHART:

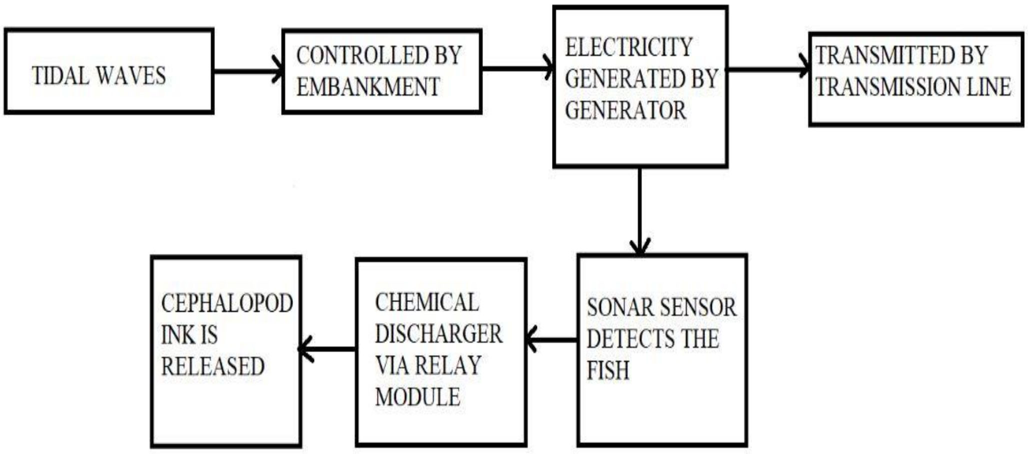

Fig.4 Process Flowchart of marine life protection system

5 ADDITIONAL DETIALS

5.1 WORKING PROCESS:

A tidal energy works via a turbine works like a wind turbine, with blades rotating 12-to-18 times a minute depending on tide strength. The turbine is connected to a gearbox that turns a generator, creating electricity. Consecutively, with the help of SONAR SENSORS we will keep an eye on the marine animals' safety. The sensor will detect the presence of fishes within a range of 11 metres. The sensor will direct this signal to the Chemical Discharger through the Relay Module. The chemicals used in the discharger are either Cephalopod Ink or Disturbances Cues. The chemical used might be subject to its availability, eco-friendliness and synthesising cost. Thus, the chemical is released which alerts the fish about the danger and hence save their lives.

5.2 SOFTWARE REQUIREMNT:

5.2.1 KEIL C

Keil MDK is the complete software development environment for a range of Arm Cortex-M based microcontroller devices. MDK includes the µVision IDE and debugger, Arm C/C++ compiler, and essential middleware components.

Fig.5.1 Keil C

5.2.2 ARDUINO IDE

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them. The Arduino IDE is similar to the C programming. The program is uploaded to the port and transferred to the Arduino Uno R3. The following picture represents the Arduino IDE logo:

Fig.5.2 Arduino IDE

5.3 REFERENCES

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Date:

Authorization letter for participation in "The Inventors Challenge 2023"

To whom so ever it may concern

Subject: Authorization of Participants for "The Inventors Challenge 2023" jointly organized by All India Council for Technical Education (AICTE), Arm Education and ST Microelectronics.

I hereby certify/authorize that the below listed faculty and students are enrolled in our institution (Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology.

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1. DR. R GOWRI SHANKAR RAO	Professor	gshankarrao@veltech.edu.in	DEPARTMENT OF PHYSICS

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1. Marthala Bhargavi	B.Tech	II /3rd Sem.	vtu23371@veltech.edu.in	Computer Science and Engineering
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Name of the idea: MARINE LIFE PROTECTION USING BIO-CHEMICAL SENSORS (TIDAL ENERGY)

Abstract of the Idea:

Tidal energy is a growing and the cleanest form of energy. This energy source can be highly advantageous as it is eco friendly and gives high output of energy. Though, it has not been implemented in a large scale due to a numerous factor such as high maintenance, geographical conditions including risk for marine life. In our project, we will develop a urea based sensor which will release 'DISTURBANCES CUES' a signal(chemicals) which marine animal will detect and will stay away from the tidal plant. This device can be put into the water body where the tidal plant is setup. This project is an attempt to understand the response mechanism of aquatic life and also solve one of the major issues hindering the setup of tidal plant.

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Signature:

14 AUG 2023

Registrar: 14/08/2023
Prof. Dr. E. Kannan
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Date: 14-08-2023